Industrial Vision

Lecture 16 - Estimating the Fundamental Matrix

The eight-point and seven-point algorithms, and RANSAC

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Where we are

In the last lecture we derived $F = K_R^{-\top}[\mathbf{t}]_\times R\, K_L^{-1}$ together with its calibrated counterpart $E = [\mathbf{t}]_\times R$, and established that the **epipoles span the null spaces** of F, since $F\mathbf{e} = \mathbf{0}$ and $F^\top \mathbf{e}' = \mathbf{0}$.

Today we treat three topics.

1. As an immediate consequence of the fact that the epipole is the null vector of F, we determine the geometry of **two special configurations**, namely parallel cameras and a forward-moving camera.
2. We estimate F from point correspondences by means of the **eight-point algorithm**, explaining why eight points are needed, how the **rank-2** condition is enforced, and why **normalization** is required.
3. We then supply the **minimal solver** that the count of seven degrees of freedom, and place it inside **RANSAC**, which is how F is estimated in practice.

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The first special case: parallel cameras

In a **non-converging** configuration the baseline lies along the image x-axis and there is no rotation. For the purposes of the algebra we take $K_L = K_R = K = \mathrm{diag}(f, f, 1), R = I$ and $\mathbf{t} = (t_x, 0, 0)^\top$.

$$[\mathbf{t}]_\times = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -t_x \\ 0 & t_x & 0 \end{bmatrix} \xrightarrow{K^{-\top}[\mathbf{t}]_\times K^{-1}} F \propto \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

The epipolar line of the point $\mathbf{x} = (x, y, 1)$ is

$$\ell' = F\mathbf{x} = (0, -1, y)^\top \;\Rightarrow\; y' = y.$$

The lines are **horizontal**, so the match lies in the **same row**. The stereo search is thereby reduced to one dimension along a scanline.

Where is the epipole? Since $F(1,0,0)^\top = \mathbf{0}$ we have

$$\mathbf{e} = (1, 0, 0)^\top, \qquad \text{a point at infinity.}$$

The epipole lies **at infinity**, as it must for parallel cameras, since the baseline never meets the image plane. The null space thus determines the configuration.

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The second special case: a forward-translating camera

The camera now moves **along the optical axis**, so that $R = I$ and $\mathbf{t} = (0, 0, t_z)^\top$. The optical axes are now collinear, and the baseline points straight into the image.

$$[\mathbf{t}]_\times = \begin{bmatrix} 0 & -t_z & 0 \\ t_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{K^{-\top}[\mathbf{t}]_\times K^{-1}} F \propto \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The epipolar line of the point $\mathbf{x} = (x, y, 1)$ is

$$\ell' = F\mathbf{x} = (-y,\ x,\ 0)^\top.$$

The constant term is zero, so every line **passes through the origin** and the lines are therefore **radial**.

Where is the epipole? Since $F(0, 0, 1)^\top = \mathbf{0}$ we have

$\mathbf{e} = (0, 0, 1)^\top$, the origin, known as the **focus of expansion**.

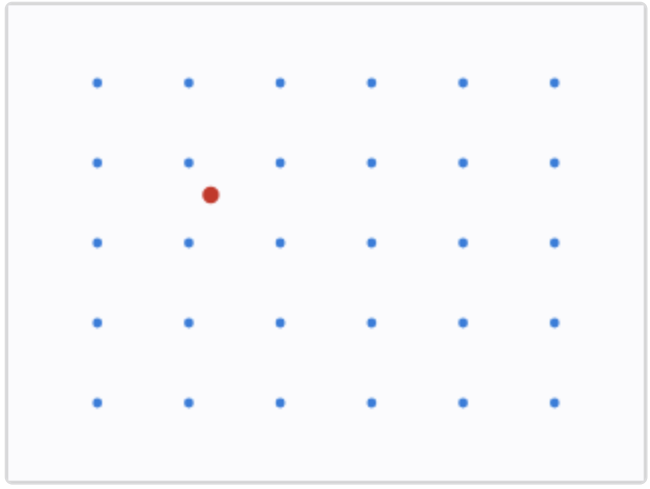
As the camera moves forward, the scene appears to stream radially outward from this point, which produces the familiar sensation of moving into the scene. The same use of the null space applies, for a different configuration.

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Demonstration of the three configurations

Parallel | Forward | Converging

left view — x (move cursor)



right view — epipolar pencil $l' = F x$



R = I, t = (t_x,0,0): F = [[0,0,0],[0,0,−1],[0,1,0]] — lines are horizontal, $y' = y$, epipole at ∞.

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In practice F is not known

Every derivation so far has assumed F to be given. In practice we have **two images and nothing else**. The correspondences must therefore come from a general **feature matcher**, such as SIFT or SURF, because a search along epipolar lines is not yet possible, that search itself requiring F.

The one relation available to us is the **epipolar constraint**, and it has the essential property of being **linear in the entries of F**:

$$\tilde{\mathbf{x}}'^{\top} F\, \tilde{\mathbf{x}} = 0 \iff \begin{bmatrix} x'x,\ x'y,\ x',\ y'x,\ y'y,\ y',\ x,\ y,\ 1 \end{bmatrix} \mathbf{f} = 0,$$

Here $\mathbf{f} = (f_{11}, f_{12}, f_{13}, f_{21}, f_{22}, f_{23}, f_{31}, f_{32}, f_{33})^{\top}$ collects the entries of F row by row.

Each correspondence contributes **one row**. This is the same construction, in which a data matrix multiplies a vector of unknowns, that was used for the homography DLT, with one additional step at the end.

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The linear system, and why eight points are required

We stack m correspondences into a data matrix A of size $m \times 9$ and solve

$$A\,\mathbf{f} = \mathbf{0}.$$

Why eight points? The matrix F has nine entries but is defined **only up to scale**, and therefore has **eight degrees of freedom**. Eight correspondences give A **rank 8**, hence a **one-dimensional null space**, which determines $\mathbf{f}$ up to scale.

$$A_{8\times 9}\,\mathbf{f} = \mathbf{0}, \qquad \mathbf{f} = \operatorname{null}(A).$$

In practice $m \gg 8$ points are used. Since the correspondences are noisy, $A\mathbf{f} = \mathbf{0}$ has no exact solution and we take the **least-squares** solution:

$$\mathbf{f} = \arg\min_{\|\mathbf{f}\|=1} \|A\mathbf{f}\| = \text{right singular vector of } A \text{ for } \sigma_{\min}.$$

This is the same use of the singular value decomposition as in every other estimator in this course.

The matrix F in fact has only **seven degrees of freedom**, since the rank-2 condition removes one further parameter. The eight-point method disregards that condition while solving the linear system and then enforces it afterwards, as described on the next slide.

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Enforcing the rank-2 condition

The vector obtained from $\mathrm{null}(A)$ is reshaped into a 3×3 matrix F, but in the presence of noise this matrix has **full rank 3**, so that $\det F \neq 0$. A genuine fundamental matrix must have **rank 2**, with $\det F = 0$, so that all the epipolar lines meet at a single epipole.

The correction. We seek the nearest rank-2 matrix $\hat{F}$ in the Frobenius norm. By the **Eckart and Young theorem** this is obtained by taking the singular value decomposition and setting the smallest singular value to zero:

$$F = U\,\mathrm{diag}(\sigma_1, \sigma_2, \sigma_3)V^\top, \qquad \boxed{\hat{F} = U\,\mathrm{diag}(\sigma_1, \sigma_2, 0)V^\top}$$

No other rank-2 matrix is closer, since $\|F - \hat{F}\|_F = \sigma_3$ is exactly the minimum attainable. By construction $\det \hat{F} = 0$.

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Normalization

See Hartley and Zisserman, §11.2

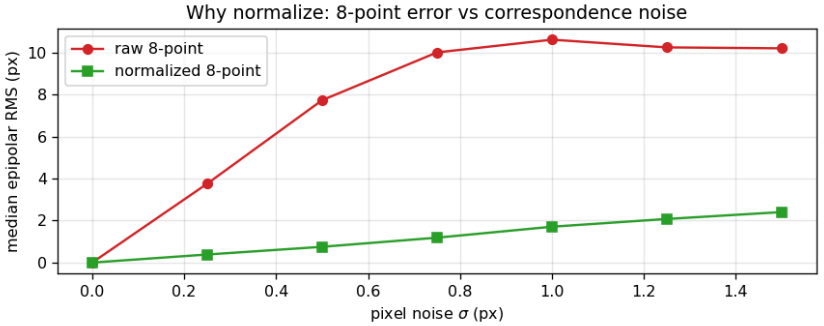
Raw pixel coordinates are **badly conditioned**, since entries of A such as $x'x$, of order 10^5, appear alongside a column of ones. The singular value decomposition is then dominated by these differences of scale, and noise seriously degrades the estimate.

Hartley's remedy is to condition each view by a similarity transformation T, which translates the centroid to the origin and scales the points so that their mean distance from it is $\sqrt{2}$.

$$\tilde{\mathbf{x}}_n = T\tilde{\mathbf{x}}, \quad \tilde{\mathbf{x}}'_n = T'\tilde{\mathbf{x}}'$$

We solve for $\tilde{F}$ using the conditioned points and then **undo** the conditioning:

$$F = T'^{\top} \tilde{F}\, T.$$



From F back to depth

Once F is known the argument closes. We may search **along the epipolar lines**, obtain dense correspondences, and from them recover depth.

For a parallel configuration with known baseline b, that

$$d = x - x' = \frac{fb}{Z} \;\Rightarrow\; Z = \frac{fb}{d}.$$

The disparity is inversely proportional to the depth

That result assumed the baseline to be known. Suppose instead that a camera is simply moved about and **the pose is not given**.

The depth Z and the translation $\mathbf{t}$ then trade off against one another, since the same disparity is consistent with a distant point and a large movement, or with a near point and a small movement. Depth is therefore recoverable only **up to an unknown scale**.

This situation is the subject of **structure from motion**, treated next.

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Extra-slides

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The seven-point algorithm, the minimal solver for F

F has **seven** degrees of freedom, being nine entries less one for the overall scale and less one for the constraint $\det F = 0$. Each correspondence supplies one equation, so **seven correspondences ought to suffice**. The eighth point of the previous method is the price of ignoring $\det F = 0$, exactly as the extra three points of DLT-PnP were the price of ignoring $R^\top R = I$. A linear method relaxes a constraint and pays for the relaxation in data.

With seven correspondences the coefficient matrix is $A \in \mathbb{R}^{7\times 9}$ of rank seven, so its null space is **two-dimensional**. Let $\mathbf{f}_1$ and $\mathbf{f}_2$ span it, with corresponding matrices F_1 and F_2. Every candidate lies in that span, and since the overall scale is free it may be written with a single parameter:

$$F(\alpha) = \alpha F_1 + (1-\alpha) F_2.$$

We now impose the constraint the eight-point method discarded:

$$\det\big(\alpha F_1 + (1-\alpha) F_2\big) = 0.$$

The determinant of a 3×3 matrix is cubic in its entries and the entries are linear in α, so this is a **cubic in α**. A real cubic has **one or three real roots**, giving one or three candidates, each of which satisfies $\det F = 0$ **exactly**. No singular value is truncated afterwards.

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What a candidate F should be scored by

A candidate must be scored against every correspondence. The obvious quantity $\tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}}$ is the **algebraic** residual, and it is not a distance: it changes with the arbitrary scale of the homogeneous coordinates, and with the arbitrary scale of F, neither of which changes the geometry. The geometric quantity is the **distance from the point to its epipolar line**.

The perpendicular distance from a point to the line $\ell = (a, b, c)^{\top}$ is $|\ell^{\top} \tilde{\mathbf{x}}| / \sqrt{a^2 + b^2}$. Applying this in the right image, where $\ell' = F\tilde{\mathbf{x}}$,

$$d\big(\tilde{\mathbf{x}}', F\tilde{\mathbf{x}}\big)^2 = \frac{\big(\tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}}\big)^2}{(F\tilde{\mathbf{x}})_1^2 + (F\tilde{\mathbf{x}})_2^2}.$$

This charges all the error to the right image, although both points are noisy. The **Sampson distance** charges it to both, and it is the usual score in modern implementations:

$$d^2_{\text{Sampson}} = \frac{\big(\tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}}\big)^2}{(F\tilde{\mathbf{x}})_1^2 + (F\tilde{\mathbf{x}})_2^2 + (F^{\top}\tilde{\mathbf{x}}')_1^2 + (F^{\top}\tilde{\mathbf{x}}')_2^2}.$$

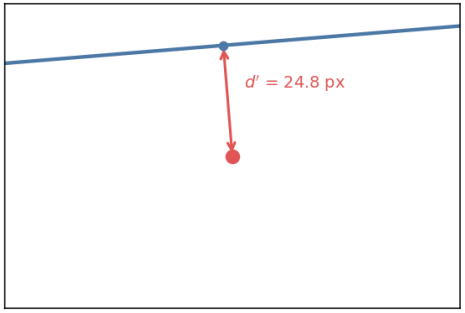
How to read the denominator. Moving $\mathbf{x}'$ changes the residual $c = \tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}}$ at the rate $\big((F\tilde{\mathbf{x}})_1, (F\tilde{\mathbf{x}})_2\big)$, and moving $\mathbf{x}$ at the rate $\big((F^{\top}\tilde{\mathbf{x}}')_1, (F^{\top}\tilde{\mathbf{x}}')_2\big)$. The denominator is the squared length of that four-vector, so the ratio is a squared distance in pixels: how far the points must move, to first order, before the residual reaches zero.

It no longer depends on the scale of F. Multiplying F by k multiplies the numerator and the denominator alike by k^2, while the algebraic residual changes by k.

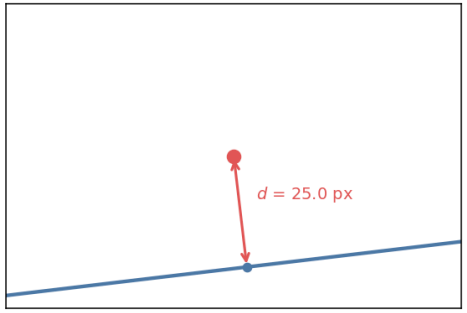
Minimizing it, rather than only using it to score, is done by the Levenberg-Marquardt of **L22**.

The residual $\mathbf{x}'^{\top} F \mathbf{x}$ is charged to both images, and the denominator $J J^{\top}$ converts it to pixels

right image: $\ell' = F\mathbf{x}$



left image: $\ell = F^{\top}\mathbf{x}'$

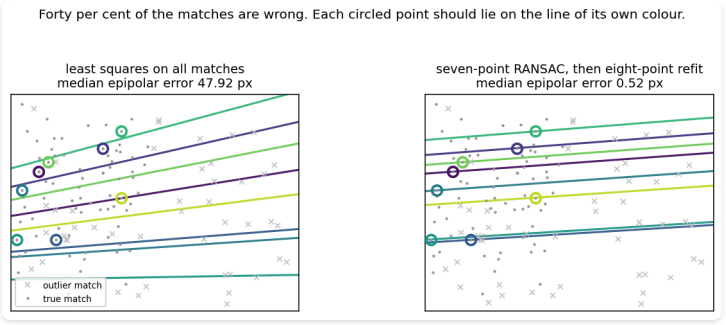


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RANSAC on F

Correspondences from SIFT always contain wrong matches, and one wrong match moves a least-squares fit arbitrarily far. In practice F is therefore estimated by **RANSAC over the minimal solver**.

1. **Sample** seven correspondences.
2. **Solve** by the seven-point algorithm, obtaining one or three candidates.
3. **Score** every correspondence against each candidate by the Sampson distance.
4. **Keep** the candidate with the largest consensus set.
5. **Refit** by the normalized eight-point algorithm on all the inliers.
6. **Refine** nonlinearly, by Levenberg-Marquardt, which **L22** develops.



The sample size sits in an exponent of $N = \log(1-p)/\log(1-w^s)$: at $w = 0.5$ and $p = 0.99$, a sample of seven needs 588 trials where one of eight needs 1177.

Questions

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Summary

- Since the **epipole is the null vector of** F, the special configurations may be read off directly. For **parallel** cameras $F \propto \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$, the lines are horizontal and $\mathbf{e}$ lies at infinity. For **forward** motion $F \propto \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, the lines are radial and $\mathbf{e}$ is the focus of expansion.
- In the **eight-point algorithm** the constraint $\tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}} = 0$ is **linear** in $\mathbf{f}$, each match contributing the row $[x'x, x'y, x', y'x, y'y, y', x, y, 1]$. The rows are stacked into A and $\mathbf{f}$ is taken as the null vector of A, that is, the singular vector belonging to the smallest singular value. Eight points are required because F has eight degrees of freedom up to scale.
- The **rank-2 condition** is imposed afterwards. The raw estimate has rank 3, and it is projected onto the nearest rank-2 matrix by the Eckart and Young theorem, setting σ_3 to zero so that $\hat{F} = U \operatorname{diag}(\sigma_1, \sigma_2, 0) V^{\top}$ and $\det \hat{F} = 0$.
- The points should be **normalized**, by moving the centroid to the origin and scaling the mean distance to $\sqrt{2}$, after which the system is solved and the conditioning removed by $F = T'^{\top} \tilde{F} T$. This reduces the error under noise by roughly a factor of ten.
- The **seven-point algorithm** is the minimal solver. Seven correspondences leave a two-dimensional null space, so the candidates form the pencil $F(\alpha) = \alpha F_1 + (1 - \alpha) F_2$, and $\det F(\alpha) = 0$ is a **cubic** with one or three real roots, each of rank 2 exactly.
- In practice F is estimated by **RANSAC** over that solver, scored by the **Sampson distance**, a squared distance in pixels, and refitted by the eight-point method on the inliers. The nonlinear refinement is the Levenberg-Marquardt of **L22**.
- F is **not determined** by a coplanar scene or by a pure rotation, both explained by a homography instead.
- The relation $Z = fb/d$ from L17 recovers **depth**, provided that the pose is known.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 11, "Computation of the Fundamental Matrix F"**, in particular §11.1 on the linear solution, §11.1.1 on the rank-2 constraint and §11.2 on normalization.

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